TEDAC HID

Button Assignment Reference

MCU: STM32H723ZGT6 (LQFP144) · Board: EC Buying FK723M1-ZGT6 V1.0

76 physical + 2 virtual buttons + 6 analog axes · USB HID Custom — 22-byte report

TDU	LHG	RHG	Total
21 buttons	28 buttons	27 buttons	76+2 buttons
BTN 01–21	BTN 22–47, 73–74	BTN 48–72, 75–76	+ 6 analog axes

★ Highlighted rows mark newly added buttons. Active-low logic: GND = button pressed. Internal pull-up on all GPIO inputs. 78 button bits (76 physical + 2 virtual) + 2-bit padding = 10 bytes.

Analog Axes (ADC1 — Circular DMA)

6 axes · signed 16-bit · –32 768 to +32 767 | Potentiometers: 3V3 → wiper → GND

Axis #	Function	STM32 Pin	Port	ADC Ch / Pin
1	Joystick X	PA6	GPIOA	CH3 / 6
2	Joystick Y	PA7	GPIOA	CH7 / 7
3	Joystick Z	PA2	GPIOA	CH14 / 2
4	Rotation X	PA3	GPIOA	CH15 / 3
5	Rotation Y	PC0	GPIOC	CH10 / 0
6	Rotation Z	PC1	GPIOC	CH11 / 1

TDU — Tactical Display Unit

21 buttons · BTN 01–21 · GPIOE (PE0–PE15) + GPIOD (PD8–PD12)

HID #	Name / Function	STM32 Pin	Port	Pin #
01	TDU TAD	PE0	GPIOE	0
02	TDU FCR	PE1	GPIOE	1
03	TDU PNV	PE2	GPIOE	2
04	TDU GS	PE3	GPIOE	3
05	TDU AZ	PE4	GPIOE	4
06	TDU SW6	PE5	GPIOE	5
07	TDU SW7	PE6	GPIOE	6
08	TDU LMC	PE7	GPIOE	7
09	TDU CAGE	PE8	GPIOE	8
10	TDU RF UP	PE9	GPIOE	9
11	TDU RF DOWN	PE10	GPIOE	10
12	TDU EL UP	PE11	GPIOE	11
13	TDU EL DOWN	PE12	GPIOE	12
14	TDU SYM UP	PE13	GPIOE	13
15	TDU SYM DOWN	PE14	GPIOE	14
16	TDU BRT UP	PE15	GPIOE	15
17	TDU BRT DOWN	PD8	GPIOD	8
18	TDU COM UP	PD9	GPIOD	9
19	TDU COM DOWN	PD10	GPIOD	10
20	TDU DAY	PD11	GPIOD	11
21	TDU NT	PD12	GPIOD	12

LHG — Left Hand Grip

28 buttons · BTN 22–47, 73–74 · GPIOD + GPIOB + GPIOC · ★ = trigger (new)

HID #	Name / Function	STM32 Pin	Port	Pin #
22	LHG SW1	PD0	GPIOD	0
23	LHG SW2	PD1	GPIOD	1
24	LHG SW3	PD2	GPIOD	2
25	LHG SW4	PD3	GPIOD	3
26	LHG SW5	PD4	GPIOD	4
27	LHG HAT1 UP	PD5	GPIOD	5
28	LHG HAT1 DOWN	PD6	GPIOD	6
29	LHG HAT1 LEFT	PD7	GPIOD	7
30	LHG HAT1 RIGHT	PB0	GPIOB	0
31	LHG HAT2 UP	PB1	GPIOB	1
32	LHG HAT2 DOWN	PB2	GPIOB	2
33	LHG HAT2 LEFT	PB3	GPIOB	3
34	LHG HAT2 RIGHT	PB4	GPIOB	4
35	LHG HAT3 UP	PB5	GPIOB	5
36	LHG HAT3 DOWN	PB6	GPIOB	6
37	LHG HAT3 LEFT	PB7	GPIOB	7
38	LHG HAT3 RIGHT	PB8	GPIOB	8
39	LHG TEMP1 A	PB9	GPIOB	9
40	LHG TEMP1 B	PB10	GPIOB	10
41	LHG TEMP2 A	PB11	GPIOB	11
42	LHG TEMP2 B	PB12	GPIOB	12
43	LHG TEMP3 A	PB13	GPIOB	13
44	LHG TEMP3 B	PC4	GPIOC	4
45	LHG 3POS A	PC5	GPIOC	5
46	LHG 3POS B	PC6	GPIOC	6
47	LHG PUSH	PC7	GPIOC	7
73	LHG TRIG POS1 ★	PD13	GPIOD	13
74	LHG TRIG POS2 ★	PD14	GPIOD	14

★ BTN 73–74 (LHG TRIG POS1 / POS2) are the two detents of the LHG trigger. They share GPIOD with the existing LHG block and were added after the initial design.

RHG — Right Hand Grip

27 buttons · BTN 48–72, 75–76 · GPIOB + GPIOC + GPIOF + GPIOG

HID #	Name / Function	STM32 Pin	Port	Pin #
48	RHG SW1	PB14	GPIOB	14
49	RHG SW2	PB15	GPIOB	15
50	RHG SW3	PC10	GPIOC	10
51	RHG SW4	PC11	GPIOC	11
52	RHG SW5	PC12	GPIOC	12
53	RHG SW6	PC13	GPIOC	13
54	RHG SW7	PF0	GPIOF	0
55	RHG HAT1 UP	PF1	GPIOF	1
56	RHG HAT1 DOWN	PF2	GPIOF	2
57	RHG HAT1 LEFT	PF3	GPIOF	3
58	RHG HAT1 RIGHT	PF4	GPIOF	4
59	RHG HAT2 UP	PF5	GPIOF	5
60	RHG HAT2 DOWN	PF6	GPIOF	6
61	RHG HAT2 LEFT	PG6	GPIOG	6
62	RHG HAT2 RIGHT	PG7	GPIOG	7
63	RHG TEMP A	PG8	GPIOG	8
64	RHG TEMP B	PG9	GPIOG	9
65	RHG 3POS1 A	PG10	GPIOG	10
66	RHG 3POS1 B	PG11	GPIOG	11
67	RHG 3POS2 A	PG12	GPIOG	12
68	RHG 3POS2 B	PG13	GPIOG	13
69	RHG PUSH	PG14	GPIOG	14
70	TDU OFF	PG15	GPIOG	15
71	SPARE 1	PF7	GPIOF	7
72	SPARE 2	PF8	GPIOF	8
75	RHG SW8	PG0	GPIOG	0
76	RHG SW9	PG1	GPIOG	1

BTN 70 (TDU OFF) maps to the Off position of the TDU Day/NT/Off 3-position switch. BTN 71–72 are spare/reserve inputs. BTN 75–76 (RHG SW8/SW9) are additional normal buttons on PG0/PG1.

Virtual Buttons

BTN 77–78 · Computed in firmware — no additional GPIO

HID #	Name	GPIO Pin	Port	Source pins (active-low)
77	Virtual A	—	GPIOF	PF3 (BTN57) + PF4 (BTN58)
78	Virtual B	—	GPIOD	PD2 (BTN24) + PD4 (BTN26)

Virtual A (BTN 77): ON = bit 1 when PF3 AND PF4 are both NOT pressed. virtual_A = !(PF3_pressed || PF4_pressed). Virtual B (BTN 78): ON = bit 1 when PD2 AND PD4 are both NOT pressed. virtual_B = !(PD2_pressed || PD4_pressed). Physical buttons (BTN 01–76) are unaffected.

GPIO Port Summary

All button inputs use GPIO_MODE_INPUT + GPIO_PULLUP

Port	Pins Used	Function
GPIOA	PA2, PA3, PA6, PA7, PA11, PA12	ADC Z/Rx axes, ADC X/Y axes, USB DP/DM
GPIOB	PB0–PB15	LHG BTN 30–43, RHG BTN 48–49
GPIOC	PC0, PC1, PC4–PC7, PC10–PC13	ADC Ry/Rz axes, LHG BTN 44–47, RHG BTN 50–53
GPIOD	PD0–PD12, PD13–PD14	TDU BTN 17–21, LHG BTN 22–29, LHG BTN 73–74
GPIOE	PE0–PE15	TDU BTN 01–16
GPIOF	PF0–PF8	RHG BTN 54–60, Spare BTN 71–72
GPIOG	PG0–PG1, PG6–PG15	RHG BTN 61–70, RHG BTN 75–76

USB HID Report Format

Descriptor: 55 bytes · Report: 22 bytes · No Report ID

Bytes	Content	Notes	Size
Bytes 0–11	6 analog axes	2 bytes each, signed 16-bit, little-endian	**12 bytes**
Bytes 12–21	76 physical + 2 virtual buttons + 2-bit padding	1 bit per button, padded to 10 bytes	**10 bytes**
Total	—	No Report ID	**22 bytes**

Byte layout: [0–11] 6×2-byte axes (signed 16-bit LE) + [12–21] 76 button bits (LSB first) + 4 padding bits. Button reading: GPIO_PIN_RESET (LOW) = pressed = bit set to 1.